INCH-POUND

MIL-DTL-24308/7G
19 July 2000
SUPERSEDING
MIL-PRF-24308/7F
2 December 1996

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRIC, RECTANGULAR, MINIATURE, POLARIZED SHELL, RACK AND PANEL, PIN CONTACTS, NONMAGNETIC, CLASSES M AND N, SOLDER TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24308.

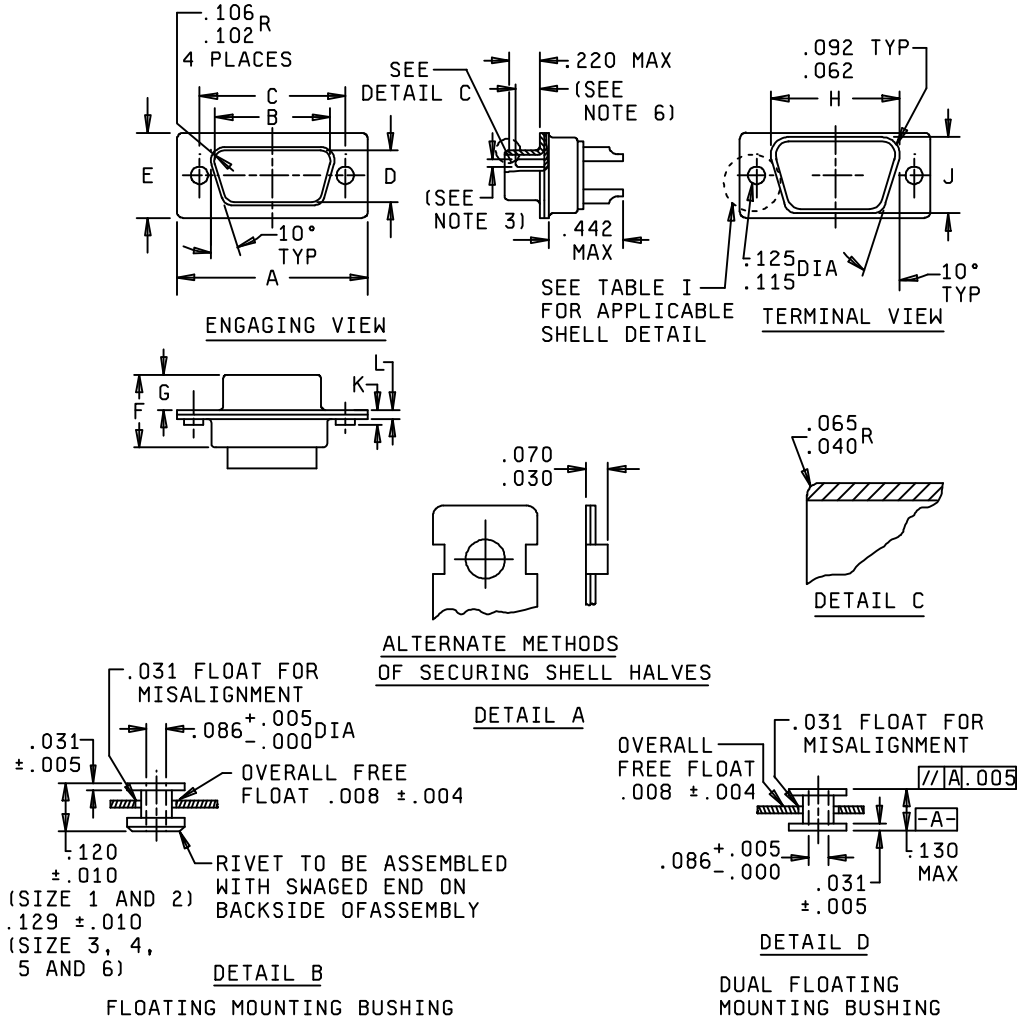

FIGURE 1. Shell, plug, solder type.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:06Z
Check the source to verify that this is the current version before use.

Detail 1/	Shell size	Dimensions										
		A	B	C	D	E	F	G	H	J	K	L
A	1	1.228 (31.19) 1.198 (30.43)	.671 (17.04) .661 (16.79)	.989 (25.12) .979 (24.87)	.334 (8.48) .324 (8.23)	.509 (12.93) .479 (12.17)	.432 (10.97) .412 (10.46)	.238 (6.05) .229 (5.82)	.769 (19.53) .749 (19.02)	.432 (10.97) .412 (10.46)	.060 (1.52) .035 (0.89)	.040 (1.02) .020 (0.51)
	2	1.556 (39.52) 1.526 (38.76)	.999 (25.37) .989 (25.12)	1.317 (33.45) 1.307 (33.20)	.334 (8.48) .324 (8.23)	.509 (12.93) .479 (12.17)	.432 (10.97) .412 (10.46)	.238 (6.05) .229 (5.82)	1.093 (27.76) 1.073 (27.25)	.432 (10.97) .412 (10.46)	.060 (1.52) .035 (0.89)	.040 (1.02) .020 (0.51)
	3	2.103 (53.42) 2.073 (52.65)	1.539 (39.09) 1.529 (38.84)	1.857 (47.17) 1.847 (46.91)	.334 (8.48) .324 (8.23)	.509 (12.93) .479 (12.17)	.436 (11.07) .416 (10.57)	.238 (6.05) .229 (5.82)	1.635 (41.53) 1.615 (41.02)	.432 (10.97) .412 (10.46)	.070 (1.78) .050 (1.27)	.049 (1.25) .029 (0.74)
	4	2.744 (69.76) 2.714 (68.94)	2.187 (55.55) 2.177 (55.30)	2.505 (63.63) 2.495 (63.37)	.334 (8.48) .324 (8.23)	.509 (12.93) .479 (12.17)	.436 (11.07) .416 (10.57)	.238 (6.05) .229 (5.82)	2.282 (57.96) 2.262 (57.45)	.432 (10.97) .412 (10.46)	.070 (1.78) .050 (1.27)	.049 (1.25) .029 (0.74)
	5	2.650 (67.31) 2.620 (66.55)	2.084 (52.93) 2.074 (52.68)	2.411 (61.24) 2.401 (60.99)	.446 (11.33) .436 (11.07)	.620 (15.75) .590 (14.99)	.436 (11.07) .416 (10.57)	.238 (6.05) .229 (5.82)	2.188 (55.58) 2.168 (55.07)	.544 (13.82) .524 (13.31)	.070 (1.78) .050 (1.27)	.049 (1.25) .029 (0.74)
	6	2.744 (69.70) 2.714 (68.94)	2.217 (56.31) 2.207 (56.06)	2.505 (63.63) 2.495 (63.37)	.508 (12.90) .498 (12.65)	.683 (17.35) .653 (16.59)	.436 (11.07) .416 (10.57)	.238 (6.05) .229 (5.82)	2.312 (58.72) 2.292 (58.22)	.606 (15.39) .586 (14.88)	.070 (1.78) .050 (1.27)	.049 (1.25) .029 (0.74)
B 2/	1	Same as detail A, shell size 1 except in accordance with detail B.										
	2	Same as detail A, shell size 2 except in accordance with detail B.										
	3	Same as detail A, shell size 3 except in accordance with detail B.										
	4	Same as detail A, shell size 4 except in accordance with detail B.										
	5	Same as detail A, shell size 5 except in accordance with detail B.										
	6	Same as detail A, shell size 6 except in accordance with detail B.										
D	1	Same as detail A, shell size 1 except in accordance with detail D.										
	2	Same as detail A, shell size 2 except in accordance with detail D.										
	3	Same as detail A, shell size 3 except in accordance with detail D.										
	4	Same as detail A, shell size 4 except in accordance with detail D.										
	5	Same as detail A, shell size 5 except in accordance with detail D.										
	6	Same as detail A, shell size 6 except in accordance with detail D.										

1/ See table I for supersession data.
2/ Detail B is inactive for new design, use detail D.

FIGURE 1. Shell, plug, solder type.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:06Z
Check the source to verify that this is the current version before use.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are in parentheses and are for general information only.
3. Pin diameter .041/.039 (1.04/0.99 mm) for arrangements A-1-1, A-2-1, A-3-1, A-4-1, A-5-1. Pin diameter .0305/.0295 (0.775/0.749 mm) for arrangements A-1-2, A-2-2, A-3-2, A-4-2, A-5-2 and A-6-1.
4. Patent notice. The Government has a royalty free license under the U.S. Patents listed below for the benefit of manufacturers of the items disclosed and described herein, either for the Government or for use in equipment delivered to the Government.

Patent numbers

2790153 expires April 23, 1974
3110093 expires November 12, 1980
3158424 expires November 24, 1981

5. Copyright notice. All information disclosed in this military specification which is of may be copyrighted by ITT Cannon Electric is reproduced herein with the express permission of the copyright owner.
6. Dimensions are .150 (3.81 mm) minimum full pin diameter extension (sizes 3 through 6) and .159 (4.04 mm) minimum full pin diameter extension (sizes 1 and 2).

Inches	mm	Inches	mm	Inches	mm
.004	0.10	.062	1.57	.115	2.92
.005	0.13	.065	1.65	.120	3.05
.008	0.20	.070	1.78	.125	3.18
.010	0.25	.086	2.18	.129	3.28
.030	0.76	.092	2.34	.130	3.30
.031	0.79	.102	2.59	.220	5.59
.040	1.02	.106	2.69	.422	10.72

FIGURE 1. Shell, plug, solder type - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:06Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

The connectors shall conform to figure 1 and the specification Part or Identifying Number (PIN) indicated in table I.

TABLE I. Design standards. 1/

Part of Identifying Number (PIN) class N	PIN class M	Shell size	Shell Detail (see figure 1)	Insert arrangement reference number 3/	Contact PIN 4/
M24308/7-1	M24308/7-23	1	A	A-1-1	B2
M24308/7-2	M24308/7-24	2	A	A-2-1	B2
M24308/7-3	M24308/7-25	3	A	A-3-1	B2
M24308/7-4	M24308/7-26	4	A	A-4-1	B2
M24308/7-5	M24308/7-27	5	A	A-5-1	B2
2/ M24308/7-12	2/ M24308/7-34	1	B	A-1-1	B2
M24308/7-13	M24308/7-35	2	B	A-2-1	B2
M24308/7-14	M24308/7-36	3	B	A-3-1	B2
M24308/7-15	M24308/7-37	4	B	A-4-1	B2
M24308/7-16	M24308/7-38	5	B	A-5-1	B2
M24308/7-17	M24308/7-39	1	D	A-2-1	B2
M24308/7-18	M24308/7-40	2	D	A-2-1	B2
M24308/7-19	M24308/7-41	3	D	A-3-1	B2
M24308/7-20	M24308/7-42	4	D	A-4-1	B2
M24308/7-21	M24308/7-43	5	D	A-5-1	B2

1/ These connectors mate with connectors specified in MIL-DTL-24308/5, MIL-DTL-24308/6.
2/ PINs M24308/7-12 through M24308/7-16 and M24308/7-34 through M24308/7-38 are inactive for new design, use M24308/7-17 through M24308/7-21 and M24308/7-39 through M24308/7-43.
3/ For insert arrangement see MIL-DTL-24308 Appendix A.
4/ For contact see MIL-DTL-24308 Appendix B.

PIN: Consists of the letter M, the basic number of the specification sheet, and a dash number compiled from the code.

PIN example M24308/7- -21 F

Specification sheet number

Data sheet number from table I

Finish: For class N:
F = Cadmium.
P = Passivated stainless steel.
No suffix = Zinc.
For class M: Class M connectors have a gold finish, no suffix is required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:06Z
Check the source to verify that this is the current version before use.

Supersession data: See table II.

TABLE II. Shell supersession data.

Shell detail (see figure 1)	Superseded MS sheet
Detail A	MS18268-1 through -6
Detail B	MS18268-7 through -12
Detail D	MS18268-13 through -18

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AV, AT, CR-4, MI
Navy - AS, CG, MC, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project: 5935-4249-04)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:06Z
Check the source to verify that this is the current version before use.